

Fundamental Analysis: Earnings & Ratios



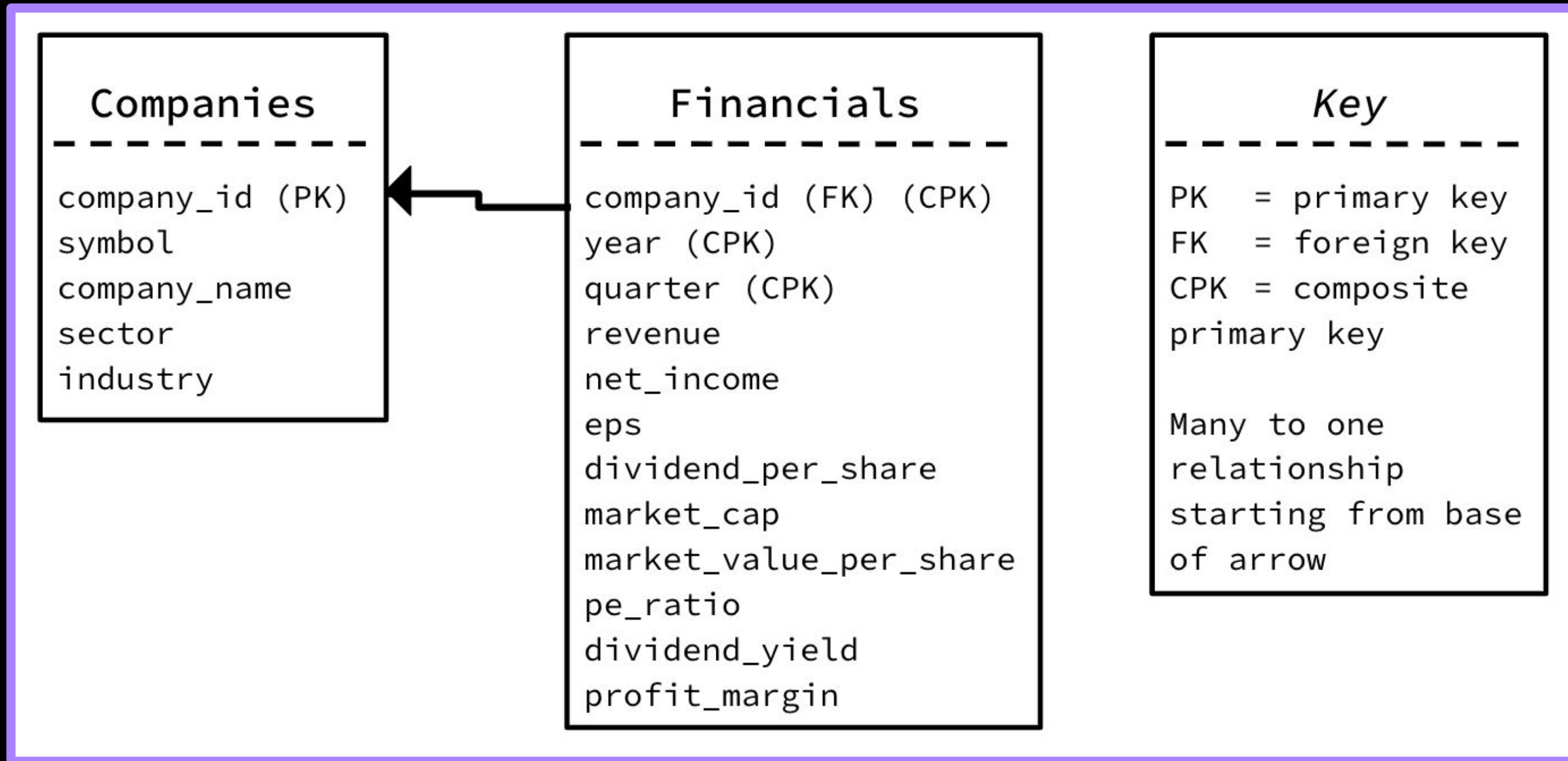
Table Of Contents

{01}	Entity Relationship Diagram	{04}	SQL Queries
{02}	Ratio Calculations	{05}	Investment Recommendations
{03}	Table Population	{06}	Extra



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{01} Entity Relationship Diagram



{02} Ratio Calculations

Formulas for calculated values

- $\text{P/E Ratio} = \text{Market Cap} / \text{Net Income}$
- $\text{Dividend Yield} = \text{Dividend Per Share} / \text{Market Value Per Share}$
- $\text{Profit Margin} = \text{Net Income} / \text{Revenue}$



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{03} Table Population - 1/6

SQL Schema Creation Script:

```
CREATE DATABASE `fundamental_analysis` /*!40100 DEFAULT CHARACTER SET  
utf8mb4 COLLATE utf8mb4_0900_ai_ci */ /*!80016 DEFAULT ENCRYPTION='N'  
*/;
```



{03} Table Population - 2/6

Creating Companies Table:

```
CREATE TABLE companies (  
    company_id INT PRIMARY KEY AUTO_INCREMENT,  
    symbol VARCHAR(50) UNIQUE NOT NULL,  
    company_name VARCHAR(100) UNIQUE NOT NULL,  
    sector VARCHAR(100),  
    industry VARCHAR(100)  
);
```



{03} Table Population – 3/6

Inserting Into Companies Table:

```
INSERT INTO companies (symbol, company_name, sector, industry) VALUES
('AAPL', 'Apple Inc.', 'Technology', 'Consumer Electronics'),
('MSFT', 'Microsoft Corporation', 'Technology', 'Software'),
('NVDA', 'NVIDIA Corporation', 'Technology', 'Semiconductors'),
('AMZN', 'Amazon.com, Inc.', 'Consumer Discretionary', 'E-Commerce'),
('META', 'Meta Platforms, Inc.', 'Technology', 'Social Media'),
('GOOGL', 'Alphabet Inc. (Class A)', 'Technology', 'Internet Services'),
('TSLA', 'Tesla, Inc.', 'Consumer Discretionary', 'Automobiles'),
('AVGO', 'Broadcom Inc.', 'Technology', 'Semiconductors'),
('GOOG', 'Alphabet Inc. (Class C)', 'Technology', 'Internet Services'),
('BRK-B', 'Berkshire Hathaway Inc. (Class B)', 'Financials', 'Conglomerate');
```



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{03} Table Population - 3/6

Inserting Into Companies Table:

	company_id	symbol	company_name	sector	industry
▶	1	AAPL	Apple Inc.	Technology	Consumer Electronics
	2	MSFT	Microsoft Corporation	Technology	Software
	3	NVDA	NVIDIA Corporation	Technology	Semiconductors
	4	AMZN	Amazon.com, Inc.	Consumer Discretionary	E-Commerce
	5	META	Meta Platforms, Inc.	Technology	Social Media
	6	GOOGL	Alphabet Inc. (Class A)	Technology	Internet Services
	7	TSLA	Tesla, Inc.	Consumer Discretionary	Automobiles
	8	AVGO	Broadcom Inc.	Technology	Semiconductors
	9	GOOG	Alphabet Inc. (Class C)	Technology	Internet Services
	10	BRK-B	Berkshire Hathaway Inc. (Class B)	Financials	Conglomerate



<Group 1>

<Project 2>



{03} Table Population - 4/6

Creating Financials Table :

```
CREATE TABLE financials (  
    company_id INT,  
    year INT NOT NULL,  
    quarter VARCHAR(2) NOT NULL,  
    revenue DECIMAL(20, 2),  
    net_income DECIMAL(20, 2),  
    eps DECIMAL(10, 2),  
    dividend_per_share DECIMAL(10, 2),  
    market_cap DECIMAL(20, 2),  
    market_value_per_share DECIMAL(10, 2),  
    dividend_yield DECIMAL(10, 2),  
    pe_ratio DECIMAL(10, 2),  
    profit_margin DECIMAL(10, 2),  
    PRIMARY KEY (company_id, year, quarter),  
    FOREIGN KEY (company_id) REFERENCES Companies(company_id)  
);
```



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{03} Table Population - 5/6

Inserting Into Financials Table:

Importing Data into Financials Table :

```
LOAD DATA LOCAL INFILE '/path/to/file.csv'  
INTO TABLE financials  
FIELDS TERMINATED BY ','  
OPTIONALLY ENCLOSED BY '"'  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS;
```



<Group 1>

<Project 2>



{03} Table Population – 6/6

Inserting Into Financials Table:

company_id	year	quarter	revenue	net_income	eps	dividend_per_share	market_cap	market_value_per_share	dividend_yield	pe_ratio	profit_margin
1	2024	an	38160000000.00	9374000000.00	6.11	0.99	329000000000.00	216.24	NULL	NULL	0.00
1	2024	Q1	90753000.00	23636000.00	1.53	0.24	265000000000.00	170.67	NULL	NULL	0.00
1	2024	Q2	85777000.00	21448000.00	1.40	0.25	323000000000.00	209.91	NULL	NULL	0.00
1	2024	Q3	94930000.00	14736000.00	0.97	0.25	352000000000.00	232.49	NULL	NULL	0.00
1	2024	Q4	124300000.00	36330000.00	2.41	0.25	377000000000.00	251.92	NULL	NULL	0.00
2	2024	an	236600000000.00	88136000000.00	12.41	3.16	320000000000.00	428.61	NULL	NULL	0.00
2	2024	Q1	61858000.00	21939000.00	2.95	0.75	313000000000.00	417.53	NULL	NULL	0.00
2	2024	Q2	64727000.00	22036000.00	2.96	0.75	332000000000.00	444.36	NULL	NULL	0.00
2	2024	Q3	65585000.00	24667000.00	3.32	0.75	320000000000.00	428.58	NULL	NULL	0.00
2	2024	Q4	69632000.00	24108000.00	3.24	0.83	313000000000.00	423.98	NULL	NULL	0.00
3	2024	an	60900000000.00	29760000000.00	5.16	0.03	244000000000.00	118.18	NULL	NULL	0.00
3	2024	Q1	22103000.00	12285000.00	NULL	NULL	152000000000.00	90.33	NULL	NULL	0.00
3	2024	Q2	26044000.00	14881000.00	0.60	0.01	213000000000.00	123.51	NULL	NULL	0.00
3	2024	Q3	30040000.00	16599000.00	0.68	0.01	287000000000.00	121.42	NULL	NULL	0.00
3	2024	Q4	35082000.00	19309000.00	0.79	0.01	325000000000.00	137.48	NULL	NULL	0.00

company_id	year	quarter	revenue	net_income	eps	dividend_per_share	market_cap	market_value_per_share	dividend_yield	pe_ratio	profit_margin
4	2024	an	590700000000.00	59248000000.00	5.53	NULL	204000000000.00	195.31	NULL	NULL	0.00
4	2024	Q1	143313000.00	10431000.00	1.00	NULL	188000000000.00	180.38	NULL	NULL	0.00
4	2024	Q2	147977000.00	13485000.00	1.29	NULL	201000000000.00	193.25	NULL	NULL	0.00
4	2024	Q3	158877000.00	15328000.00	1.46	NULL	196000000000.00	186.33	NULL	NULL	0.00
4	2024	Q4	187792000.00	20004000.00	NULL	NULL	232000000000.00	221.30	NULL	NULL	0.00
5	2024	an	142700000000.00	62360000000.00	23.86	2.00	136000000000.00	537.22	NULL	NULL	0.00
5	2024	Q1	36455000.00	12369000.00	4.86	0.50	124000000000.00	483.81	NULL	NULL	0.00
5	2024	Q2	39071000.00	13465000.00	5.31	0.50	128000000000.00	502.89	NULL	NULL	0.00
5	2024	Q3	40589000.00	15688000.00	6.20	0.50	144000000000.00	571.47	NULL	NULL	0.00
5	2024	Q4	48385000.00	20838000.00	NULL	0.50	148000000000.00	590.71	NULL	NULL	0.00
6	2024	an	38900000000.00	10012000000.00	8.04	0.80	213000000000.00	172.04	NULL	NULL	0.00
6	2024	Q1	80539000.00	23662000.00	1.91	0.20	188000000000.00	150.21	NULL	NULL	0.00
6	2024	Q2	84742000.00	23619000.00	1.91	0.20	226000000000.00	181.49	NULL	NULL	0.00
6	2024	Q3	88268000.00	26301000.00	2.14	0.20	204000000000.00	165.47	NULL	NULL	0.00
6	2024	Q4	96469000.00	26536000.00	NULL	0.20	232000000000.00	191.02	NULL	NULL	0.00



{03} Table Population – 6/6

Inserting Into Financials Table:

company_id	year	quarter	revenue	net_income	eps	dividend_per_share	market_cap	market_value_per_share	dividend_yield	pe_ratio	profit_margin
7	2024	an	94700000000.00	7130000000.00	2.04	NULL	833000000000.00	263.17	NULL	NULL	0.00
7	2024	Q1	21301000.00	1171000.00	0.37	NULL	559850000000.00	175.79	NULL	NULL	0.00
7	2024	Q2	25500000.00	1478000.00	0.46	NULL	631000000000.00	197.88	NULL	NULL	0.00
7	2024	Q3	25182000.00	2167000.00	0.68	NULL	840000000000.00	261.63	NULL	NULL	0.00
7	2024	Q4	25707000.00	2314000.00	NULL	NULL	1300000000000.00	417.41	NULL	NULL	0.00
8	2024	an	54530000000.00	10400000000.00	2.16	2.23	795000000000.00	174.11	NULL	NULL	0.00
8	2024	Q1	11961000.00	1325000.00	0.29	0.53	602000000000.00	130.94	NULL	NULL	0.00
8	2024	Q2	12487000.00	2121000.00	0.46	0.53	747000000000.00	159.12	NULL	NULL	0.00
8	2024	Q3	13072000.00	-1875000.00	-0.40	0.53	792000000000.00	171.52	NULL	NULL	0.00
8	2024	Q4	14054000.00	4324000.00	0.92	0.59	1040000000000.00	234.87	NULL	NULL	0.00
9	2024	an	350020000000.00	100120000000.00	8.04	0.80	2130000000000.00	173.39	NULL	NULL	0.00
9	2024	Q1	96469000.00	23662000.00	1.91	0.20	1880000000000.00	151.54	NULL	NULL	0.00
9	2024	Q2	88268000.00	23619000.00	1.91	0.20	2260000000000.00	182.76	NULL	NULL	0.00
9	2024	Q3	84742000.00	26301000.00	2.14	0.20	2040000000000.00	166.81	NULL	NULL	0.00
9	2024	Q4	80539000.00	26536000.00	NULL	0.20	2320000000000.00	192.47	NULL	NULL	0.00
10	2024	an	369000000000.00	88990000000.00	41.25	NULL	939000000000.00	434.93	NULL	NULL	NULL
10	2024	Q1	101468000.00	12702000.00	5.88	NULL	908000000000.00	420.52	NULL	NULL	0.00
10	2024	Q2	113509000.00	30348000.00	14.08	NULL	877000000000.00	406.80	NULL	NULL	0.00
10	2024	Q3	117510000.00	26251000.00	12.18	NULL	992000000000.00	460.26	NULL	NULL	0.00
10	2024	Q4	91745000.00	19694000.00	NULL	NULL	978000000000.00	452.16	NULL	NULL	0.00



{03} Table Population – 5/6

Inserting Into Financials Table:

Formula Calculations on Financials table:

```
-- calculating dividend_yield
UPDATE financials
SET dividend_yield = dividend_per_share / market_value_per_share
WHERE dividend_per_share IS NOT NULL AND market_value_per_share IS NOT NULL;
```

```
-- calculating P/E ratio
UPDATE financials
SET pe_ratio = market_cap / net_income
WHERE market_cap IS NOT NULL AND net_income IS NOT NULL;
```

```
-- calculating profit margin
UPDATE financials
SET profit_margin = net_income / revenue
WHERE net_income IS NOT NULL AND revenue IS NOT NULL;
```

Handling small numbers

(dividend_yield) by scaling:

```
UPDATE financials
SET dividend_yield = (dividend_per_share
/ market_value_per_share) * 100
WHERE dividend_per_share IS NOT NULL AND
market_value_per_share IS NOT NULL;
```



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{03} Table Population – 6/6

Inserting Ratios Data Calculations:

dividend_yield	pe_ratio	profit_margin
0.457824639289678	35.10	0.25
0.140622253471612	112117.11	0.26
0.119098661331047	150596.79	0.25
0.107531506731472	238870.79	0.16
0.099237853286758	103770.99	0.29
0.737266979305196	36.31	0.37
0.179627811175245	142668.31	0.35
0.168782068593033	150662.55	0.34
0.174996500069999	129727.98	0.38
0.195763951129770	129832.42	0.35
0.025385005923168	81.99	0.49
NULL	123728.12	0.56

dividend_yield	pe_ratio	profit_margin
0.008096510404016	143135.54	0.57
0.008235875473563	172901.98	0.55
0.007273785277859	168315.29	0.55
NULL	34.43	0.10
NULL	180232.00	0.07
NULL	149054.51	0.09
NULL	127870.56	0.10
NULL	115976.80	0.11
0.372286958787834	21.81	0.44
0.103346354974060	100250.63	0.34
0.099425321640916	95061.27	0.34
0.087493656709889	91789.90	0.39

dividend_yield	pe_ratio	profit_margin
0.087493656709889	91789.90	0.39
0.084643903099660	71024.09	0.43
0.465008137642409	21.27	2.57
0.133146927634645	79452.29	0.29
0.110198909030801	95685.68	0.28
0.120867831026772	77563.59	0.29
0.104701078421108	87428.40	0.28
NULL	1168.30	0.08
NULL	478095.64	0.05
NULL	426928.28	0.06
NULL	387632.67	0.09
NULL	561797.75	0.09

dividend_yield	pe_ratio	profit_margin
1.280799494572397	76.44	0.19
0.404765541469375	454339.62	0.11
0.333081950729010	352192.36	0.17
0.309001865671642	-422400.00	-0.14
0.251202793034445	240518.04	0.31
0.461387623277006	21.27	0.29
0.131978355549690	79452.29	0.25
0.109433136353688	95685.68	0.27
0.119896888675739	77563.59	0.31
0.103912298020471	87428.40	0.33
NULL	10.55	0.24
NULL	71484.81	0.13

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{04} SQL Queries - Approach

Query 1

The most profitable companies.



Query 2

The best dividend-paying stocks.



Query 3

Companies with the highest revenue growth.



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Query 1 – Most Profitable Companies

{  } Query

{  } Results

```
SELECT
  f.company_id,
  c.company_name,
  ROUND(SUM(f.net_income) / SUM(f.revenue) *
100, 2) AS overall_profit_margin
FROM Financials f
JOIN Companies c ON c.company_id = f.company_id
WHERE f.year = 2024
GROUP BY f.company_id, c.company_name
ORDER BY overall_profit_margin DESC;
```

company_id	company_name	overall_profit_margin
6	Alphabet Inc. (Class A)	255.91
3	NVIDIA Corporation	48.88
5	Meta Platforms, Inc.	43.69
2	Microsoft Corporation	37.25
9	Alphabet Inc. (Class C)	28.60
1	Apple Inc.	24.56
10	Berkshire Hathaway Inc. (Class B)	24.11
8	Broadcom Inc.	19.06
4	Amazon.com, Inc.	10.03
7	Tesla, Inc.	7.53



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Query 2 – Best Dividend-Paying

{  } Query

```
SELECT
  f.company_id,
  c.company_name,
  ROUND(AVG(f.dividend_yield), 2) AS
avg_dividend_yield
FROM Financials f
JOIN Companies c ON c.company_id =
f.company_id
WHERE f.year = 2024
GROUP BY f.company_id, c.company_name
ORDER BY avg_dividend_yield DESC;
```

{  } Results

company_id	company_name	avg_dividend_yield
8	Broadcom Inc.	0.52
2	Microsoft Corporation	0.29
6	Alphabet Inc. (Class A)	0.19
9	Alphabet Inc. (Class C)	0.19
1	Apple Inc.	0.18
5	Meta Platforms, Inc.	0.15
3	NVIDIA Corporation	0.01
4	Amazon.com, Inc.	NULL
7	Tesla, Inc.	NULL
10	Berkshire Hathaway Inc. (Class B)	NULL



Query 3 — Companies with the highest revenue growth.

{  } Query

{  } Results

```
SELECT
  q1.company_id,
  c.company_name,
  ROUND(((q4.revenue - q1.revenue) / q1.revenue) * 100,
2) AS revenue_growth_percent
FROM Financials q1
JOIN Financials q4
  ON q1.company_id = q4.company_id
  AND q1.year = q4.year
  AND q1.quarter = 'Q1'
  AND q4.quarter = 'Q4'
JOIN Companies c ON c.company_id = q1.company_id
ORDER BY revenue_growth_percent DESC;
```

company_id	company_name	revenue_growth_percent
3	NVIDIA Corporation	58.72
1	Apple Inc.	36.97
5	Meta Platforms, Inc.	32.73
4	Amazon.com, Inc.	31.04
7	Tesla, Inc.	20.68
6	Alphabet Inc. (Class A)	19.78
8	Broadcom Inc.	17.50
2	Microsoft Corporation	12.57
10	Berkshire Hathaway Inc. (Class B)	-9.58
9	Alphabet Inc. (Class C)	-16.51



<Group 1>

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{05} Investment Recommendations

Advice:

Invest in Alphabet Inc. Class A!

Not only is it at the top in our other query for most profitable companies but it is also a dividend paying stock. Dividend paying stocks tend to be more stable and are generally safer to invest in.



{06} Extra Improvements - 1/4

Summary of Table Improvements:

- Removed 'report_id' column from Financials Table based on advice from previous project: COMPOSITE PRIMARY KEY OF company_id, year, quarter
- Deleted ratios table and moved columns to financials table
 - Less data repetition
- Extra columns: 'quarter', 'market_value_per_share'
- 'dividend' renamed to 'dividend_per_share' for a more descriptive name



{06} Extra Improvements – 2/4

Hypothetical Scenario – Example of using roles

Scenario: Database is being used at a company where there are many people who need different levels of access – an intern, a data updater (assuming data is updated manually), and a database admin. How can we make sure everyone is only accessing what they need to access?



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{06} Extra Improvements - 3/4

Intern

```
CREATE ROLE intern; -- create role
CREATE USER john IDENTIFIED BY 'word101'; -- create user
GRANT intern TO john; -- assign role
GRANT SELECT ON fundamental_analysis.companies TO intern;
GRANT SELECT ON fundamental_analysis.financials TO intern;
```

Data Updater

```
CREATE ROLE updater; -- create role
CREATE USER jesse IDENTIFIED BY 'hello67'; -- create user
GRANT updater TO jesse;
GRANT SELECT,UPDATE ON fundamental_analysis.companies TO updater;
GRANT SELECT,UPDATE ON fundamental_analysis.financials TO updater;
```

Database Admin

```
CREATE ROLE admin; -- create role
CREATE USER jake IDENTIFIED BY 'password123'; -- create USER
GRANT admin TO JAKE; -- assign role
GRANT ALL PRIVILEGES ON fundamental_analysis.* TO admin;
```



{06} Extra Improvements – 4/4

Optional Triggers for 3 Columns Calculated Through Equations

```
CREATE TRIGGER insert_financials
BEFORE INSERT ON financials
FOR EACH ROW
BEGIN
    SET NEW.pe_ratio = NEW.market_cap / NEW.net_income;
    SET NEW.dividend_yield = NEW.dividend_per_share / NEW.market_value_per_share;
    SET NEW.profit_margin = NEW.net_income / NEW.revenue;
END;
```

```
CREATE TRIGGER update_financials
BEFORE UPDATE ON financials
FOR EACH ROW
BEGIN
    SET NEW.pe_ratio = NEW.market_cap / NEW.net_income;
    SET NEW.dividend_yield = NEW.dividend_per_share / NEW.market_value_per_share;
    SET NEW.profit_margin = NEW.net_income / NEW.revenue;
END;
```



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Thank You!

Questions?



Credits



SlidesCarnival for the presentation template
Pexels for the photos